THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2012: Fundamental Concepts of Mathematics

May 13, 2024 9:30 a.m. – 12:00 noon

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL Questions

Note: You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

1. (10 points) Let A, B be nonempty sets, and let R, S be relations on A and B respectively. A function $f : A \longrightarrow B$ is said to be *relation preserving* if for any $x, y \in A$, we have

$$xRy \iff f(x)Sf(y).$$

Prove or disprove the following statement: For any nonempty sets A, B and for any relations R, S on A and B respectively, if $f : A \longrightarrow B$ is a surjective relation preserving function and R is an equivalence relation on A, then S is an equivalence relation on B.

The following solution is generated by an AI-based tool (with slight modification of wordings).

> "Disproof": Let $A = \{1, 2, 3\}$ and $B = \{4, 5\}$. Let R and S be the relations on A and B defined as follows:
>
> $$R = \{(1,1), (2,2), (3,3), (1,2), (2,1)\} \text{ and } S = \{(4,4), (5,5)\}.$$
>
> Consider the function $f : A \longrightarrow B$ given by $f(1) = 4$, $f(2) = 4$ and $f(3) = 5$. Then f is a surjective relation preserving function. However, we note that the relation R is an equivalence relation on A but S is not an equivalence relation on B as it is not transitive.
>
> Hence we have found a counterexample where R is an equivalence relation on A, $f : A \longrightarrow B$ is relation preserving and surjective, but S is not an equivalence relation on B.

If you think you should **prove** the given statement, point out the mistake(s) in the above "disproof" and explain briefly what goes wrong. Then provide a correct proof for the statement.

If you think you should **disprove** it, comment and find out the mistake(s) (if any) in the above disproof.

2. (30 points) **Prove or disprove** each of the following statements:

(a) For any sets A, B, C, if $A \triangle B = C \triangle B$, then $A = C$.

(b) There exist an integer m and an odd integer n such that the equation $x^2 + 2mx + 2n = 0$ has an integer solution x.

(c) There exist nonempty sets A, B, C and functions $f : A \longrightarrow B$, $g : B \longrightarrow C$ such that $(g \circ f)^{-1}$ exists but both f^{-1} and g^{-1} do not exist.

(d) For any nonempty sets A, B and for any functions $f : A \longrightarrow B$ and $g : A \longrightarrow B$, if $f^{-1}(Y) = g^{-1}(Y)$ for any subset Y of B, then $f = g$.

(e) For any subset S of the set $\mathbb{I}$ of irrational numbers, if S is infinite, then S is uncountable.

3. (16 points)

Consider the sequence $F_1, F_2, F_3, \ldots$ of *Fibonacci numbers* satisfying $F_1 = 1$, $F_2 = 1$ and

$$F_n = F_{n-1} + F_{n-2} \quad \text{for any integer } n \geq 3.$$

(a) Prove by induction that for any integer $n \geq 2$,

$$F_n^2 - F_{n+1}F_{n-1} \;=\; (-1)^{n+1}.$$

(b) Compute the first four terms of $5F_n^2 + 4(-1)^n$. What do you observe about the values obtained? Make a claim based on your observation, and provide a proof to support your claim.

4. (16 points)

(a) Prove by induction that $5^m - 4m^2 - 1 > 0$ for any integer $m \geq 2$.

(b) Find all primes p such that $5^p + 4p^4$ is a perfect square.

5. (14 points)

For any nonempty sets A, B, let $f : \mathcal{P}(A \cup B) \longrightarrow \mathcal{P}(A) \times \mathcal{P}(B)$ be defined by

$$f(X) \;=\; (X \cap A,\, X \cap B).$$

(a) Prove that f is injective.

(b) Show, by giving an example of A and B, that f is not surjective.

(c) The function f will become surjective if an assumption is imposed on the sets A and B. What is this assumption? Prove that f is surjective under this additional assumption.

6. (14 points) For each $r \in \mathbb{N}$, define the set

$$A_r = \big\{n \in \mathbb{N} : n = p_1^{k_1} \cdots p_r^{k_r} \text{ for some distinct primes } p_1, \ldots, p_r \text{ and for some } k_1, \ldots, k_r \in \mathbb{N}\big\}.$$

(a) Write down any two elements of A_2.

(b) Which of the set A_r does 2024 belong to? Explain briefly.

(c) Prove that A_r is denumerable for any $r \in \mathbb{N}$.

(d) Construct a partition P of $\mathbb{N}$ such that P is itself denumerable and each of the elements of P is also denumerable. Prove your answer.

******* **END OF PAPER** *******